

SymPy Bug Card

Bug 44: `continuous_domain` Misses Positive Periodic Intervals

Task. Find the real continuous domain of a square root with a periodic radicand.

$$\text{continuous_domain}(\sqrt{\sin(x)}, x, \mathbb{R})$$

SymPy's answer.

$$\text{continuous_domain}(\sqrt{\sin(x)}, x, \mathbb{R}) = \text{Interval}(0, \pi)$$

Correct answer.

$$\bigcup_{n \in \mathbb{Z}} [2n\pi, (2n+1)\pi]$$

Explanation. The returned set excludes $5\pi/2$, but $\sin(5\pi/2) = 1$, so $\sqrt{\sin(5\pi/2)} = 1$. The missing point lies in the translated positive interval $[2\pi, 3\pi]$, so this is not an equivalent single-period representation.

Likely root cause. The diagnosis identifies `sympy/calculus/util.py` calling `solve_univariate_inequality` for the radical constraint $\sin(x) \geq 0$; the inequality solver in `sympy/solvers/inequalities.py` solves only on one period and returns that slice without lifting it over all integer translates. This single-period root cause is a known public defect (open issue #27573, closed #9721, fixes #27720 / #27745); the specific `continuous_domain(sqrt(sin(x)))` composition appears unreported.

Artifact (GitHub).

[bug-report/bug-044-continuous-domain-misses-positive-periodic-intervals/](#)